

MULTIMODAL MEDICAL INTELLIGENCE

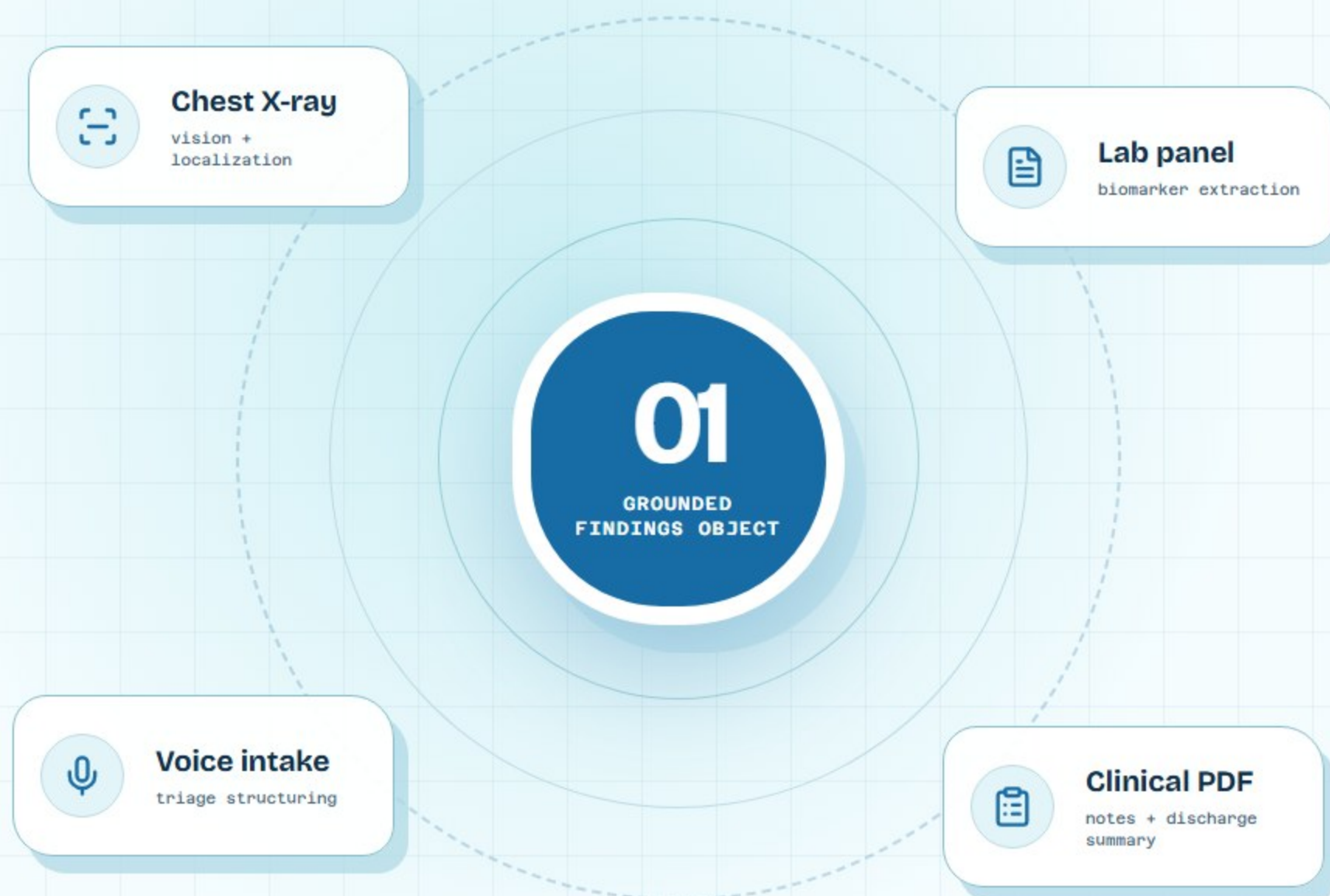
From any medical artifact to actual clinical report.

One agentic engine understands imaging, labs, documents and voice, then composes grounded outputs for physicians and patients.

Multimodal reasoning

Agentic orchestration

Malaysia-localized



Medical data is multimodal. Every handoff breaks the chain.

Four gaps stand between what clinicians upload and what patients and doctors can safely act on.

● 4 DISCONNECTED MOMENTS

01 / SEE



Imaging stays invisible

Text-first assistants cannot inspect a film or point to the anatomy behind a finding.

HANDOFF

02 / STRUCTURE



Labs stay unstructured

Hb, WBC, and renal panels get summarized as text, not stored as data you can trend.

HANDOFF

03 / GROUND



Context goes missing

Generic answers ignore patient history, Malaysia MOH guidance and local priors.

HANDOFF

04 / EXPLAIN



One voice fits nobody

Clinical language overwhelms patients; simplified output underserves doctors.

THE OPPORTUNITY

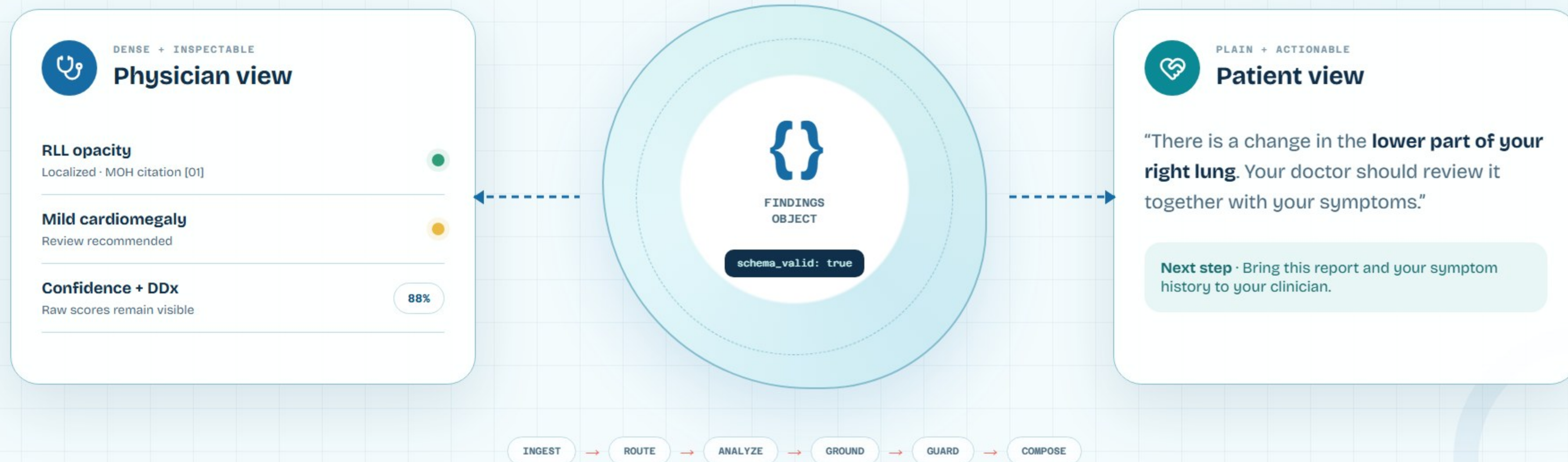
Replace four handoffs with one safe, grounded source of truth.

Built for Malaysia

One clinical truth. Two useful voices.

SihatAI routes any artifact into one canonical findings object, then adapts the explanation, not the evidence.

● NO DUPLICATED INFERENCE



The evidence stays in the room.

● CONCEPTUAL PRODUCT VIEW

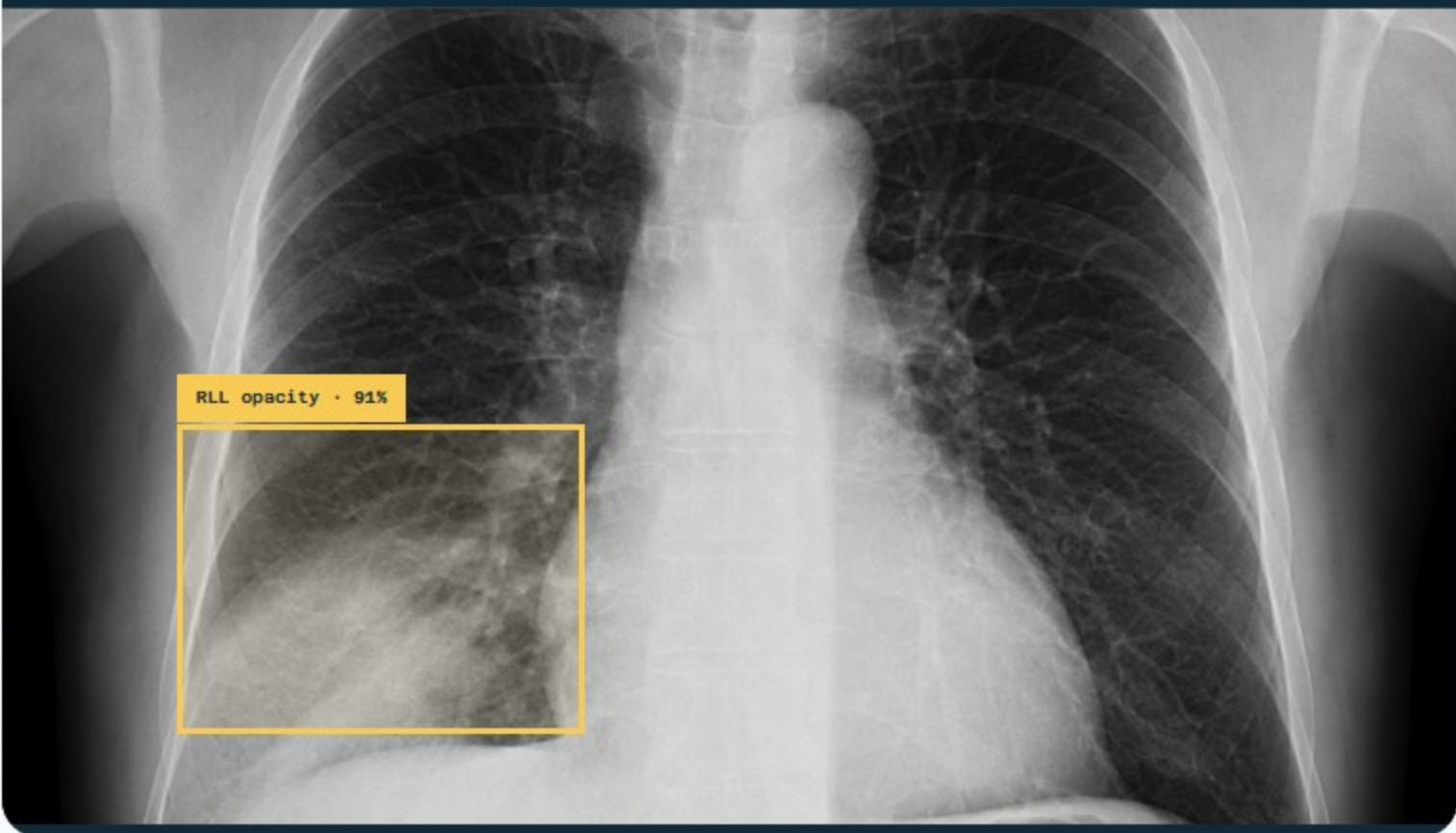
S

AR

MEDICAL RECORD / MR-2048
Chest X-ray analysis

Physician Patient

PA CHEST · 2048 × 2048 100% · OVERLAY ON



RLL opacity · 91%

OVERALL CONFIDENCE

88%

ANALYSIS COMPLETE

Right lower lobe opacity

Patchy airspace opacity localized to the right lower zone.

ABNORMAL

Mild cardiomegaly

Cardiothoracic ratio is mildly increased.

REVIEW

MOH CPG

Community Acquired Pneumonia in Adults

[01]

↳ Bounding-box localization

↳ Confidence calibration

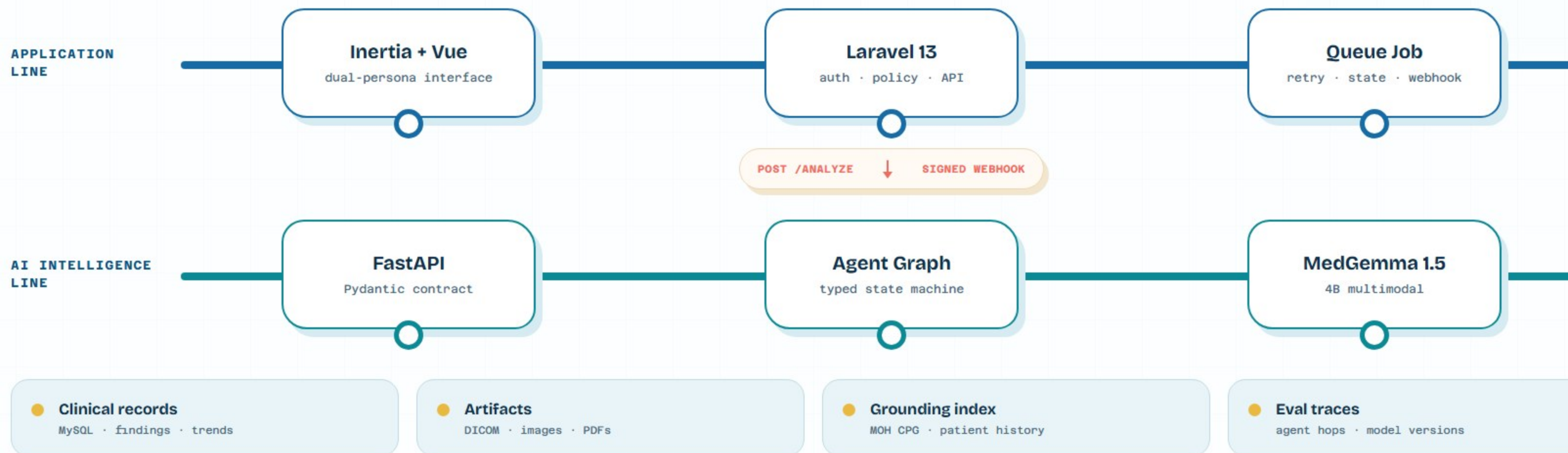
↳ Inline citations

↳ Longitudinal comparison

Laravel orchestrates. Python reasons.

A clean asynchronous contract lets product engineering and model inference scale independently.

● ONE ASYNC INTERCHANGE



Six specialists. One inspectable relay.

● BOUNDED HOPS · TYPED TOOLS

**De-ID**

scrub PII / PHI

safe_uri

**Router**

detect modality

route_confidence

**Imaging / Doc**

parallel specialists

partial_findings

**Retriever**

hybrid RAG

citations[]

**Guardrail**

veto + escalate

ALLOW | WARN

**Composer**

audience adapt

reports()

LIVE TRACE
JOB_7F3A · COMPLETE00
deidentify()
120 ms01
detect_modality()
800 ms02
medgemma_infer()
11.2 s03
rag_retrieve()
900 ms04
safety_check()
450 ms05
compose(locale=bn)
3.1 s

Not a chatbot. A clinical perception stack.

GOOGLE / MEDGEMMA-1.5-4B-IT

01 · VISION

IMAGE

See the finding

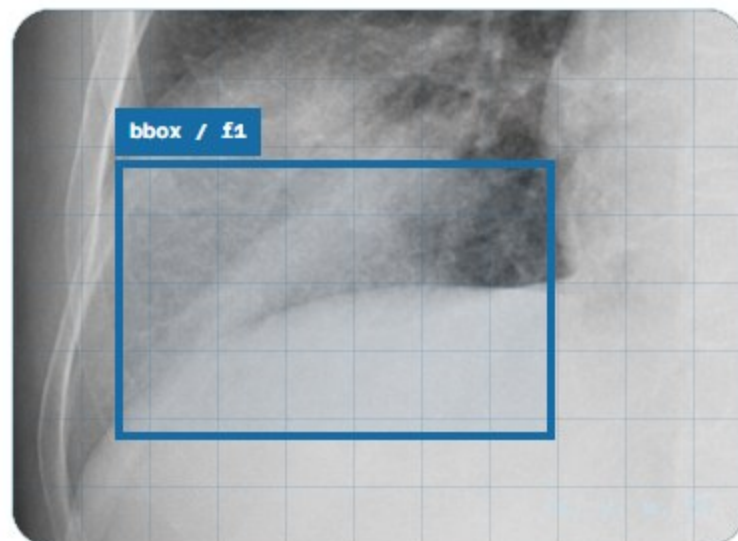


pixel tensor → structured observations[]

02 · GROUNDING

WHERE

Point to anatomy



normalized bbox → viewport-aware overlay

03 · DOCUMENT AI

DATA

Structure the PDF

Hb	9.2	LOW
Hct	28.1	LOW
RBC	3.4	LOW
WBC	3.8	LOW
Neut	1.2	LOW
PLT	85	LOW
MCV	72	LOW
CRP	48	HIGH

PDF / OCR → Pydantic schema → MySQL

04 · TEMPORAL

CHANGE

Reason over time



current ⊕ prior → new | stable | resolved

Grounded in evidence. Fluent in Malaysia.

BM

EN

中文

தமிழ்

RETRIEVAL QUERY

"RLL opacity + fever + cough"

01

Hybrid search

dense vectors + lexical BM25

02

MMR rerank

relevance × diversity

03

Confidence gate

abstain below threshold

#1	MOH CPG · Community Acquired Pneumonia	0.89
#2	Patient history · respiratory episode	0.82
#3	MOH CPG · Tuberculosis management	0.76

English

Bahasa Melayu

RINGKASAN UNTUK PESAKIT

Terdapat perubahan pada bahagian bawah paru-paru kanan.

Imej menunjukkan kawasan legap yang mungkin berkaitan dengan jangkitan. Dapatan ini perlu dinilai bersama gejala dan pemeriksaan doktor.

[MOH-CPG-01] Garis Panduan Pneumonia

89% relevance

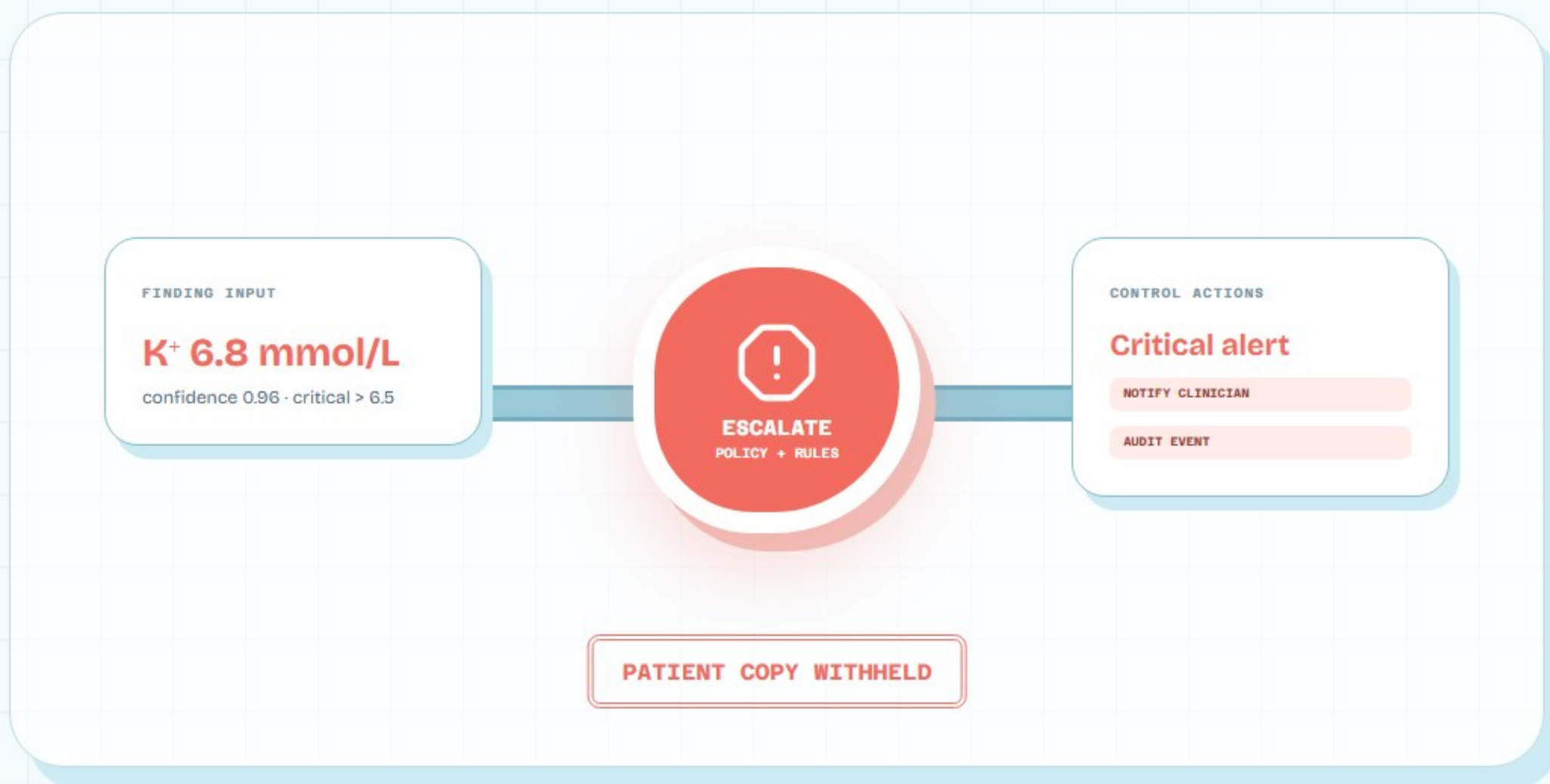
MY-LoRA

QLoRA adapter

BM medical register · local disease priors

Safety is a control plane. Not a disclaimer.

● GUARDRAIL HAS VETO POWER



01

De-identify first

DICOM tags, document PII and metadata are scrubbed before inference.

02

Calibrate uncertainty

Publish $\geq .80$ · hedge .50-.80 · abstain below .50.

03

Constrain claims

Structured DDx with confidence, never unsupported certainty.

04

Preserve oversight

Physicians inspect evidence, boxes and citations before sign-off.

We measure the system. Not just the interface.

● BENCHMARKED PIPELINE



EVAL HARNESS

Model / adapter regression loop



PRODUCTION-ORIENTED STACK

PRODUCT	Laravel 13	Inertia v3	Vue 3	MySQL	Redis
AI	FastAPI	Pydantic	MedGemma	QLoRA	RAG
MLOPS	GPU serving	adapters	trace logs	eval runs	webhooks

Our advantage is not one model. It is the system around it.

17 INTEGRATED CAPABILITIES



01

Understands more

X-rays, CT, labs, PDFs and voice all route through one pipeline. Modality detection sends each artifact to the right specialist without re-uploading.

X-RAY · CT · MRI

LAB PDFS

VOICE INTAKE



02

Grounded in Malaysia

MOH Clinical Practice Guidelines, traceable citations and a localized adapter keep every claim grounded. Reports ship in BM, EN and other local languages.

MOH CPG

CITATIONS

BM · EN · ZH · TA



03

Fails safely

De-ID runs before inference, and confidence bands decide when to publish, hedge or abstain. Guardrails can veto unsafe output and escalate critical values.

DE-ID FIRST

ABSTENTION

VETO + ESCALATE



04

One truth, two audiences

MedGemma runs once to produce a single structured findings object. That same truth becomes a physician report and a patient-friendly summary without duplicated inference.

PHYSICIAN REPORT

PATIENT REPORT

SHARED FINDINGS

SYSTEM OUTCOME CHAIN

Multimodal intake

Any clinical artifact enters one routed pipeline

Inspectable reasoning

Agent hops, traces and citations stay visible

Safer clinical action

Guardrails route uncertainty to clinicians

— THE NEXT INTERFACE IS NOT A CHATBOX

 Understands images

 Escalates safely

 Grounds every claim

 Adapts to people

It is a safe, grounded clinical intelligence layer.

SihatAI combines multimodal perception, agentic reasoning, local evidence and audience-adaptive generation in one inspectable pipeline.

01

Understands

imaging · labs · documents · voice

02

Reasons

agents · RAG · longitudinal context

03

Protects

de-ID · abstention · escalation

04

Adapts

physician · patient · multilingual